AELL101: Introduction to Electrical Engineering

Midsemester Examination

Marks: 100 + 10 (bonus); 2 hours

Instructions: *Please mention appropriate* ***units*** *against each answer for full marks.* ***No clarification will be provided. Make assumptions if you think some information is missing. Section A needs to be answered within this question paper. Section B needs to be answered in the separate answer sheet.***

Name:

Entry Number:

Section A: It contains 8 short questions in this section. Answer neatly, using only the space provided below each question. Page 4 and 5 white spaces can be used for spillover, only if needed. Use space wisely. Use separate answer sheets for rough work, if needed)

1. **(12 marks)** Answer following short questions in one or two words.

   a. The sum of currents entering a junction equals the sum of currents leaving the junction: True or False? True

   b. Impedance of a capacitor is directly proportional to the frequency: True or False? False

   c. The resistance of a 100 W, 200 V lamp is 400 ohms

   d. In a balanced three-phase AC circuit, the sum of all three generated voltages is 0 or Zero

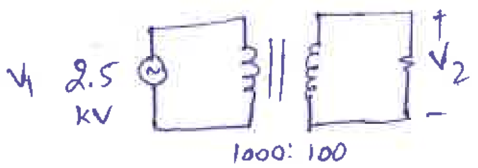
   e. If the angle between voltage and current is 30 degrees, the power factor is 0.866

   f. An RC circuit with R=10 ohms and C=200 μF will have time constant of 2 milli seconds.

   g. A half-wave rectifier has 100 V rms as input voltage. The average output DC voltage would be 45.01 V

   h. If inputs of a NAND gate are 0 and 1, the output would be 1

2. **(5 marks)** A 5-kVA single-phase transformer has 1000 and 100 turns on primary (supply side) and secondary (load side) windings, respectively. It is fed from a 2.5-kV supply. Neglecting losses, determine: (a) the secondary side voltage magnitude and (b) the primary current at rated load kVA.



$N_1 = 1000$

$N_2 = 100$

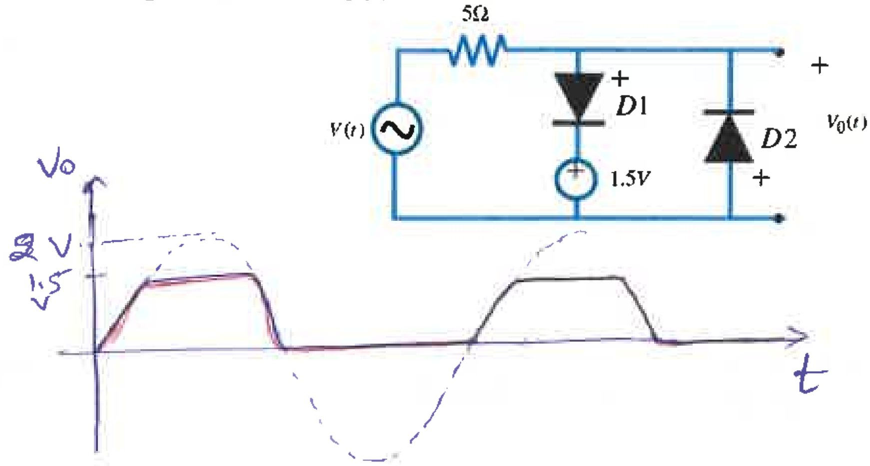
a.) $\frac{V_1}{V_2} = \frac{N_1}{N_2} \Rightarrow V_2 = \frac{V_1}{10} = 250$ Volts

b.) rated load = 5 kVA

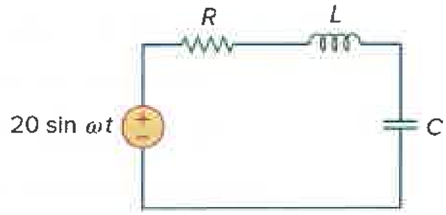
$V_1 I_1 = 5$ kVA

$\Rightarrow I_1 = \frac{5 \text{ kVA}}{2.5 \text{ kV}} = 2$ A

3. **(5 marks)** Given that $v(t) = 2\sin(wt)\,V$, as shown in the figure below. Assuming ideal diodes, sketch the output waveform $v_0(t)$.



4. **(5 marks)** In the circuit given, $R = 2\Omega$, $L = 1\text{mH}$, and $C = 0.4\mu\text{F}$. Determine, (a) Resonant frequency f_r, (b) Quality factor and (c) Bandwidth.



a.) $\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}}$

$= 0.5 \times 10^5$ rad/s

$f_r = \frac{\omega_r}{2\pi} = 7957.74$ Hz

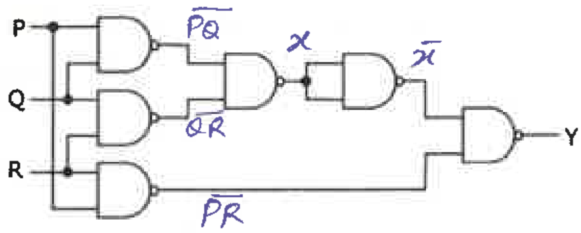
b) $Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r \cdot C \cdot R} = \frac{50 \times 10^3 \times 10^{-3}}{2} = 25$

c) Bandwidth $= \frac{\omega_r}{Q} = \frac{0.5 \times 10^5}{25} = 2 \times 10^3$ rad/sec.

5. **(5 marks)** Can we use permanent magnets to make a transformer instead of a wire-wound iron core? Justify your answer in less than 3 lines.

No. transformer works on the principle of changing flux. A permanent magnet will have a constant flux which will not induce voltage in secondary.

6. **(5 marks)** Using Boolean algebra techniques, write the expression Y as given in the circuit and simplify it as much as possible.



$$Y = \overline{\overline{x} \cdot \overline{PR}}$$

where $x = \overline{\overline{PQ} \cdot \overline{QR}}$ (use $\overline{\overline{A} \cdot \overline{B}} = A + B$)

$= PQ + QR$

$Y = \overline{\overline{PQ+QR} \cdot \overline{PR}}$ (use $\overline{\overline{A} \cdot \overline{B}} = A + B$)

$Y = PQ + QR + PR$

7. **(5 marks)** The mean length of a rectangular magnetic circuit is 40 cm, with a cross-sectional area of 8 cm², and the relative permeability of the core material is 200. When a magnetizing coil carries a current of 6 A, the magnetic flux produced in the core is 3 mWb. Calculate the following (with units):
(a) reluctance of the magnetic circuit, (b) magnetic field intensity and (c) number of turns of the coil.
(d) If a small air gap is introduced in the magnetic circuit, would more or fewer turns be required to establish the same magnetic flux? Justify your answer in 1-2 lines.

$l = 40 \times 10^{-2}$ m
$A = 8 \times 10^{-4}$ m^2
$\mu_r = 200$
$\mu_0 = 4\pi \times 10^{-7}$
$i = 6$ A
$\Phi = 3 \times 10^{-3}$ Wb

a) $\mathcal{R} = \dfrac{l}{\mu_0 \mu_r A}$

$= \dfrac{40 \times 10^{-2}}{200 \mu_0 \times 8 \times 10^{-4}}$

$= 1.989 \times 10^{6}$ A/Wb

b) $H = \dfrac{F}{l} = \dfrac{Ni}{l} = \dfrac{\Phi \mathcal{R}}{l}$

$= \dfrac{3 \times 10^{-3} \times 1.98 \times 10^{6}}{40 \times 10^{-2}}$

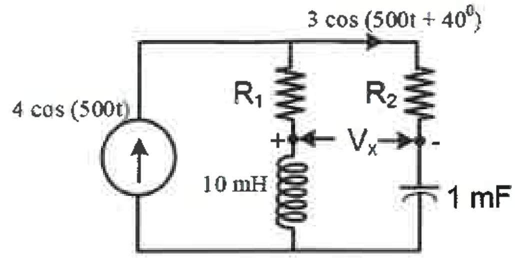
$= 14925$ A/m.

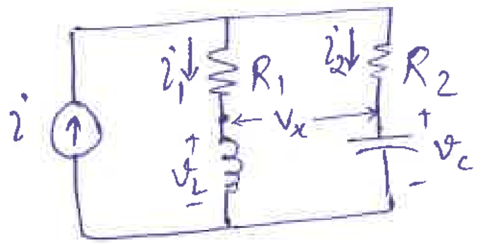
c) $F = Ni \Rightarrow N = \dfrac{F}{i} = \dfrac{Hl}{i}$

$= \dfrac{14925 \times 40 \times 10^{-2}}{6}$

$= 995$ turns

8. **(8 marks)** Find the $V_x(t)$ for the circuit shown in the figure. The current in two of the branches are given in the figure.





$$i = 4\cos(500t) = 4\angle 0^\circ$$

$$i_2 = 3\cos(500t + 40^\circ) = 3\angle 40^\circ$$

$$i_1 = i - i_2 \quad \text{(using KCL)}$$

$$= 4\angle 0^\circ - 3\angle 40^\circ$$

$$= 2.572\angle -48.57^\circ$$

$\omega = 500$;

$$V_L = i_1 \cdot j\omega L$$

$$= 2.572\angle -48.57^\circ \cdot j(5)$$

$$= 2.572\angle -48.57^\circ \cdot 5\angle 90^\circ$$

$$V_L = 12.86\angle 41.43^\circ$$

$\omega L = 500 \times 10 \times 10^{-3} = 5$

$\frac{1}{j\omega C} = \frac{1}{j500 \times 10^{-3}} = -j2$

$$V_c = i_2 \cdot \frac{1}{j\omega C}$$

$$= 3\angle 40^\circ \cdot (-2j)$$

$$= 3\angle 40^\circ \cdot 2\angle -90^\circ$$

$$V_c = 6\angle -50^\circ$$

$$V_x = V_L - V_c$$

$$= 12.86\angle 41.43^\circ - 6\angle -50^\circ$$

$$V_x = 14.326\angle 66.18^\circ$$

$$\Downarrow$$

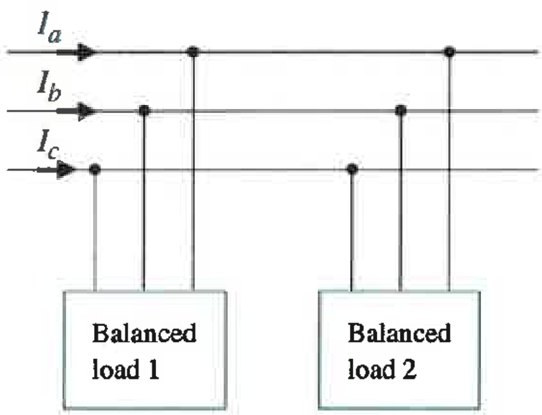
$$V_x(t) = 14.326\cos(500t + 66.18^\circ)$$

Section B: It contains 4 questions in this section. Answer neatly in the separate answer sheet provided to you. Start each question at fresh page)

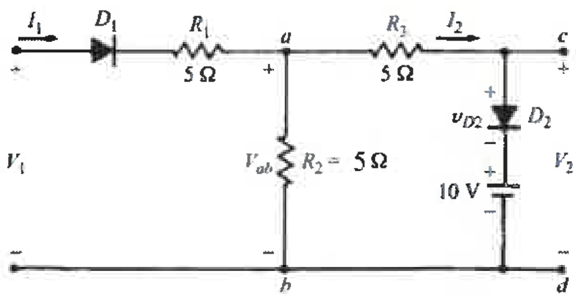
9. **(15 marks)** Two balanced loads are connected to a 240-kV rms 50-Hz line, as shown in the Figure below. Load-1 draws 30 kW at a power factor of 0.6 (lagging), while load-2 draws 45 kVAR at a power factor of 0.8 (lagging). Assuming the abc-sequence, determine:

   a) The real, reactive and complex powers absorbed by the combined load

   b) The three line currents (I_a, I_b, I_c): both magnitudes and angle



10. **(15 marks)** Find the V_2 for $-30 < V_1 < 30$ assuming both diodes are ideal. Plot V_2 as a function of V_1



11. (15 marks) You are given a 1-bit comparator module that compares two single-bit binary inputs, A and B. The module provides three outputs G, L and E:

If A > B, G is 1
If A < B, L is 1
If A = B, E is 1

a) Design this 1-bit comparator and draw circuit using NOT and AND and OR gates for all 3 outputs.

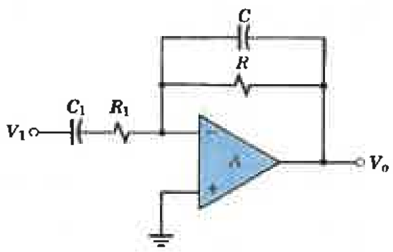
b) Now, consider two 2-bit binary numbers: $A = A_1A_0$ and $B = B_1B_0$. Using two 1-bit comparator blocks from part (a) and other required gates, design and draw a 2-bit comparator circuit that compares A and B and provides the following outputs in form 2 LEDs L1 and L2:

- If A > B , L1 lights up
- If A < B , L2 lights up
- If A = B , L1 and L2 both remains off

12. (15 marks) Your friend attended a live Beyoncé concert where she debuted a new song titled *"Echoes of Voltage"*. Excited by the performance, your friend recorded the song on their phone. However, the recording includes unwanted noise from the concert environment. To enhance the listening experience, your friend seeks your help. Since you've completed AELL101, you analyse the recording and your friend's preferences, and make the following observations:

- Most Beyoncé songs your friend enjoys have vocals cantered around 1~~00 Hz~~.
- The recording includes a low-frequency hum (~20 Hz) from the crowd and occasional high-pitched sirens (~1000 Hz) from distant ambulances.

Using an OpAmp, resistors, and capacitors available in the lab, you design the following signal processing circuit and apply the recording to the input terminal V_1. You select the component values as: $R_1 = 10\ k\Omega$, $C_1 = 1\ \mu F$, $R = 100\ k\Omega$



Now, complete the following:

a) Determine the best value of C such that the output predominantly retains Beyoncé's vocals while minimizing the noise components.

b) Calculate the critical (cut-off) frequencies of your designed device. Based on your results, briefly (in 2-4 lines) comment on the potential applications and limitations of this circuit in audio signal processing.

9.

(a) Load 1: $P_1 = 30\,kW$, $pf_1 = 0.6$ (lagging)

$\Rightarrow \cos\theta_1 = 0.6 \Rightarrow \sin\theta_1 = \sqrt{1-0.6^2} = 0.8$

$S_1 = \frac{P_1}{pf_1} = \frac{30}{0.6} = 50\ kVA$

$Q_1 = S_1 \sin\theta = 50\times0.8 = 40\ kVAR$

$S_1 = 30 + j40\ kVA$

Load 2: $Q_2 = 45\ kVAR$, $pf_2 = 0.8 = \cos\theta_2$

$\sin\theta_2 = \sqrt{1-0.8^2} = 0.6$

$|S_2| = \frac{Q_2}{\sin\theta_2} = \frac{45}{0.6} = 75\ kVA$

$P_2 = S_2 . \cos\theta_2 = 75\times0.8 = 60\,kW$

$S_2 = 60 + j45\ kVA$

Combined load: $S_1 + S_2 = \boxed{90 + 85j}$

total real power $= 90\ kW$

total reactive power $= 85\ kVAR$

total complex power $= 90 + j85\ kVA$

(b) $V_L = 240\ kV \angle 0^\circ$

method-1 $\quad \sqrt{3}|V_L||i_{L_1}| = S_1 = 50\ kVA$

Load 1 $\Rightarrow |i_{L_1}| = \frac{50\ kVA}{\sqrt{3}\times240\ kV} = 0.12\ A$

Load 2 $\Rightarrow |i_{L_2}| = \frac{75\ kVA}{\sqrt{3}\times240} = 0.18\ A$

θ_1 is angle b/w i_{P_1} & v_{P_1} (Not b/w i_{L_1} & v_{L_1})

if $\angle V_L = 0^\circ$ & if $\boxed{\text{Y load}}$: $\angle I_L = \angle i_P$

$\angle i_L = \angle i_P = \angle V_P - \theta = -30^\circ - \theta$

if load is Δ: $\angle i_L = \angle i_P - 30^\circ$

$\angle i_P = \angle V_P - \theta = \angle V_L - \theta$

$\angle i_L = -\theta - 30^\circ$

so, angle of $i_{L_1} = -\theta_1 - 30^\circ$

$= -53.13^\circ - 30^\circ$

$= -83.13^\circ$

angle of $i_{L_2} = -\theta_2 - 30^\circ$

$= -36.86 - 30^\circ$

$= -66.86^\circ$

$i_{L_1} = 0.12 \angle -83.13^\circ$

$i_{L_2} = 0.18 \angle -66.86^\circ$

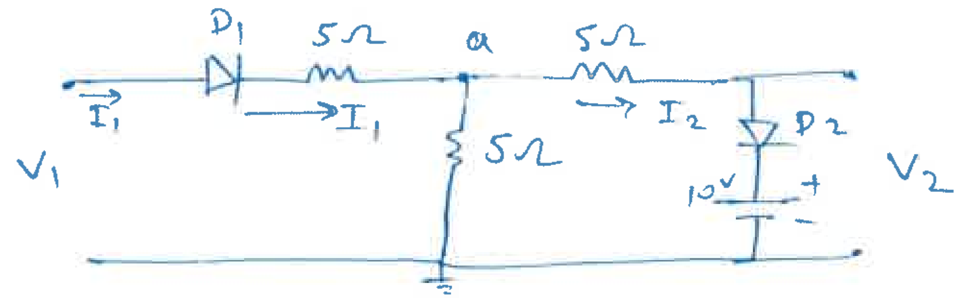
$i_a = i_{L_1} + i_{L_2}$

$= 0.29 \angle -73^\circ$

$i_b = 0.29 \angle -193^\circ$

$i_c = 0.29 \angle 47^\circ$

10)



Case I : D_1 & D_2 both ON

KCL at a : $\frac{V_1 - V_a}{5} = \frac{V_a - 10}{5} + \frac{V_a}{5}$

$\Rightarrow \quad 3V_a = V_1 + 10$

since ~~both on :- V~~

for D_2 to be ON : $V_a > 10$ Volts. $\Rightarrow \frac{V_1+10}{3} > 10$

$\Rightarrow \boxed{V_1 > 20V}$

for D_1 to be ON : $V_1 > V_a$

$\Rightarrow V_1 > \frac{V_1 + 10}{3}$

$\Rightarrow V_1 > 5$ Volts.

$\boxed{\text{so for } V_1 \geq 20 \text{ Volts. both } D_1 \text{ \& } D_2 \text{ ON} \quad V_2 = 10 \text{ Volts.}}$

Case - II : D_1 ON & D_2 OFF

$I_2 = 0 \Rightarrow V_a = V_1 \cdot \frac{5}{10} = \frac{V_1}{2}$

for D_2 to be off : $V_a < 10$ Volts.

$\Rightarrow V_1 < 20$ Volts.

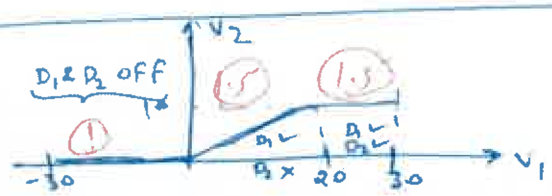
for D_1 to be ON : $V_1 > V_a \Rightarrow V_1 > 0$

$V_2 = V_{ab} = V_a = \frac{V_1}{2}$

Case - III : D_1 & D_2 both OFF : $V_1 < 0$, $V_2 = 0$

Case - IV : D_1 off & D_2 ON : (I_2 in opposite dirc not possible)

Ans:



$$V_2 = \begin{cases} 0 & V_1 < 0 \quad \text{— (1)} \\ V_1/2 & 0 < V_1 < 20 \quad \text{— (1.5)} \\ 10 & V_1 > 20 \quad \text{— (1.5)} \end{cases}$$

~~10~~)

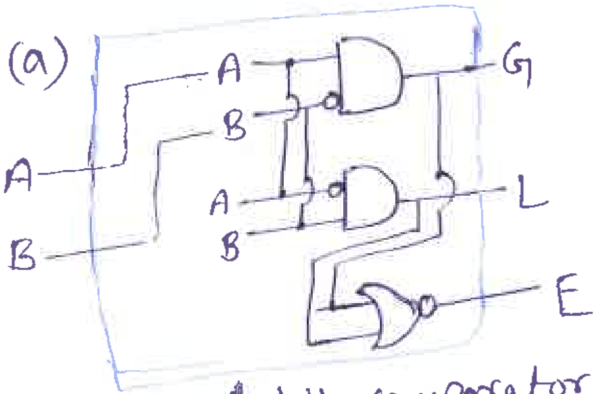
11.) 1-bit compartor

G: A>B
L: A<B
E: A=B

A	B	G	L	E
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

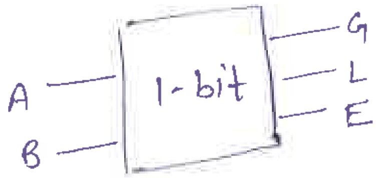
$$G = A\bar{B}$$
$$\Rightarrow \quad L = \bar{A}B$$
$$\therefore E = AB + \bar{A}\bar{B}$$
$$= \overline{(\bar{A}B + A\bar{B})}$$
$$= \overline{G + L}$$

(a)



1-bit comparator

$\Rightarrow$



(b) $A = A_1 A_0$
$B = B_1 B_0$

bit by bit comparison of a 2-bit number

bit by bit comparison & check which number has higher bit

$\Rightarrow$ start from MSB

if $A_1 > B_1 \Rightarrow L_1$ ON $= 1$ $\Rightarrow$ if $G_1 = 1 \Rightarrow L1 = 1, L2 = 0$

if $A_1 < B_1 \Rightarrow L_2$ ON $= 1$ $\Rightarrow$ if $L_1 = 1 \Rightarrow L2 = 1, L1 = 0$

if $(A_1 = B_1)$ and $(A_0 > B_0) \Rightarrow L_1$ ON $\Rightarrow$ if $(E_1 = 1$ & $G_0 = 1) \Rightarrow L1 = 1$

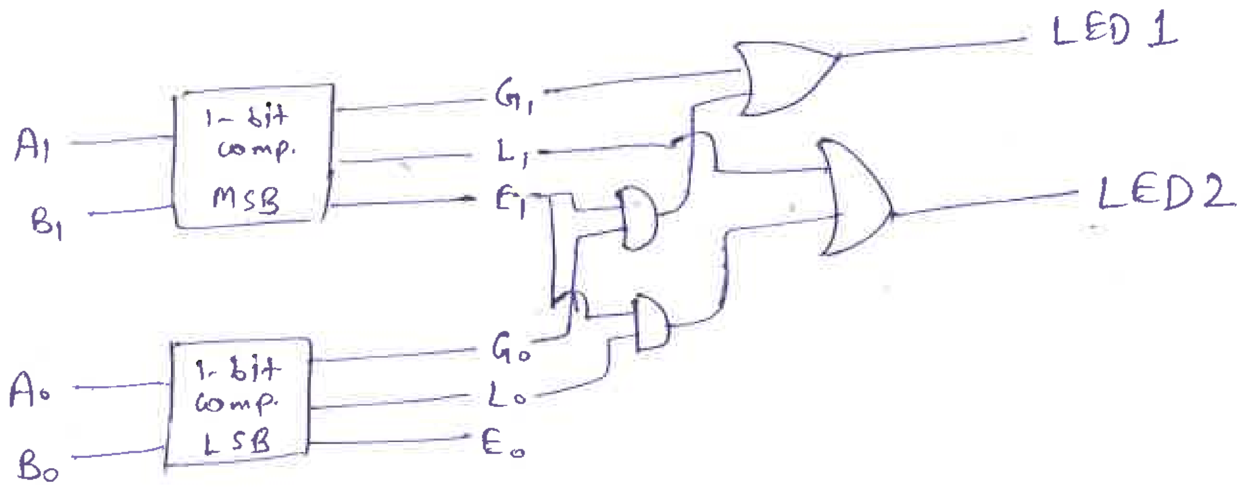
if $(A_1 = B_1)$ and $(A_0 < B_0) \Rightarrow L_2$ ON $\Rightarrow$ if $(E_1 = 1$ & $L_0 = 1) \Rightarrow L2 = 1$

if $(A_1 = B_1)$ and $(A_0 = B_0) \Rightarrow L_1$ off and L_2 off $\Rightarrow$ if $(E_1 = 1)$ & $(E_2 = 1) \Rightarrow L$

say, G_0, L_0, E_0 are output of LSB comparator $(A_0$ & $B_0)$

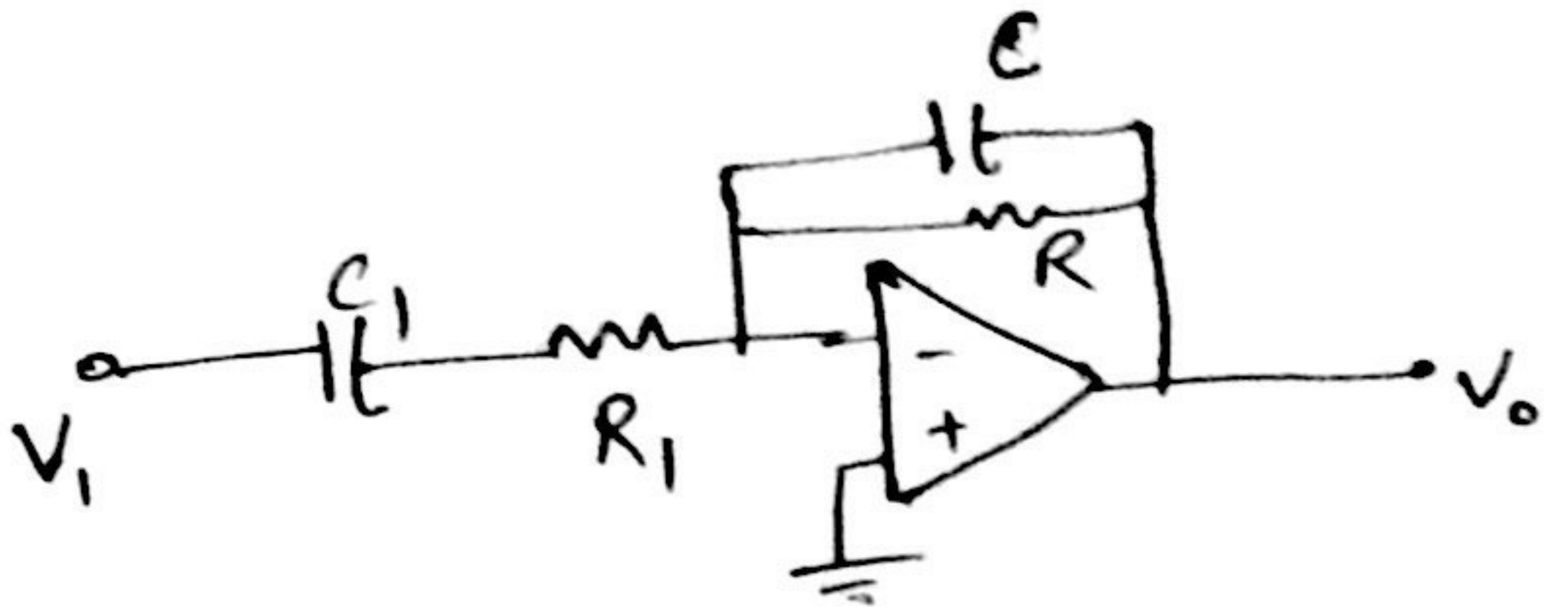
G_1, L_1, E_1 are output of MSB comparator $(A_1$ & $B_1)$

$$\Rightarrow \quad L1 = G_1 + G_0 . E_1$$
$$L2 = L_1 + L_0 . E_1$$



2-bit comparator

12)



let's solve to find gain $= \frac{V_o}{V_1} = -\frac{Z_{feedback}}{Z_{input}}$

$$\frac{V_o}{V_1} = -\frac{R \parallel \frac{1}{j\omega C}}{R_1 + j\omega C_1} = \frac{-1}{(R_1 + j\omega C_1)\left(\frac{1}{R} + j\omega C\right)}$$

(a)

$$\text{gain} = \frac{V_o}{V_1} = -\frac{1}{\left(\frac{R_1}{R} + \frac{C}{C_1}\right) + j\left(\omega R_1 C - \frac{1}{R\omega C_1}\right)}$$

$$(\text{gain})^2 = \left|\frac{V_o}{V_1}\right|^2 = -\frac{1}{\left(R_1/R + \frac{C}{C_1}\right)^2 + \left(\omega R_1 C - \frac{1}{R\omega C_1}\right)^2}$$

we want to maximize gain at 100 Hz $\Rightarrow f_r = 100$ Hz

$\Rightarrow$ gain will be maximum when denominator is min

$\Rightarrow$ max gain when $\omega R_1 C = \frac{1}{R\omega C_1} \Rightarrow \omega_r^2 = \frac{1}{RR_1CC_1}$

$$\Rightarrow C = \frac{1}{RR_1C_1 \cdot \omega_r^2} = \frac{1}{10^4 \times 10^5 \times 10^{-6} \times (2\pi 100)^2}$$

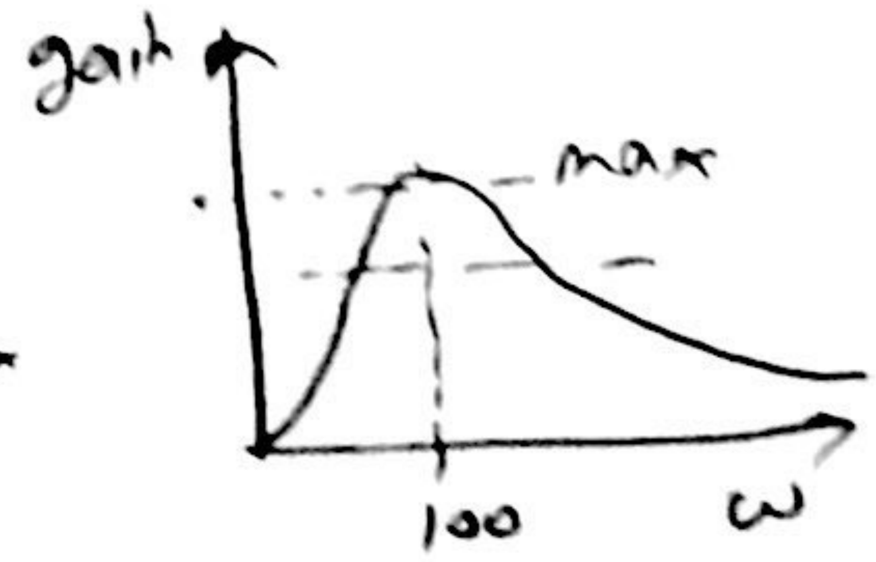
$$\Rightarrow C = 2.53 \times 10^{-9} F = \boxed{2.53\ nF} \text{ Ans.}$$

$$\text{max. gain} = -\frac{1}{R_1/R + \frac{C}{C_1}} = -\frac{1}{0.1025} = 9.75$$

b) note that when $\omega \to 0 \Rightarrow$ gain $\to 0$

$\omega \to \infty \Rightarrow$ gain $\to 0$

$\omega = 2\pi \times 100 \Rightarrow$ gain $= 9.75$ max



cut-off freq. when gain $= \frac{\text{max gain}}{\sqrt{2}}$

$$\Rightarrow \frac{(9.75)^2}{2} = \frac{1}{(0.1025)^2 + \left(2.53\times10^{-5}\omega - \frac{10}{\omega}\right)^2}$$

$$\Rightarrow (0.1025)^2 = \left(2.53\times10^{-5}\omega - \frac{10}{\omega}\right)^2$$

$$\Rightarrow 2.53\times10^{-5}\omega - \frac{10}{\omega} = \pm 0.1025$$

two cases of quadratic Eqn:

Case 1: $2.53\times10^{-5}\omega - \frac{10}{\omega} = +0.1025$

$\Rightarrow$ solve quadratic eqn for ω

$\Rightarrow$ $\omega_1 = 4143$ rad/sec.

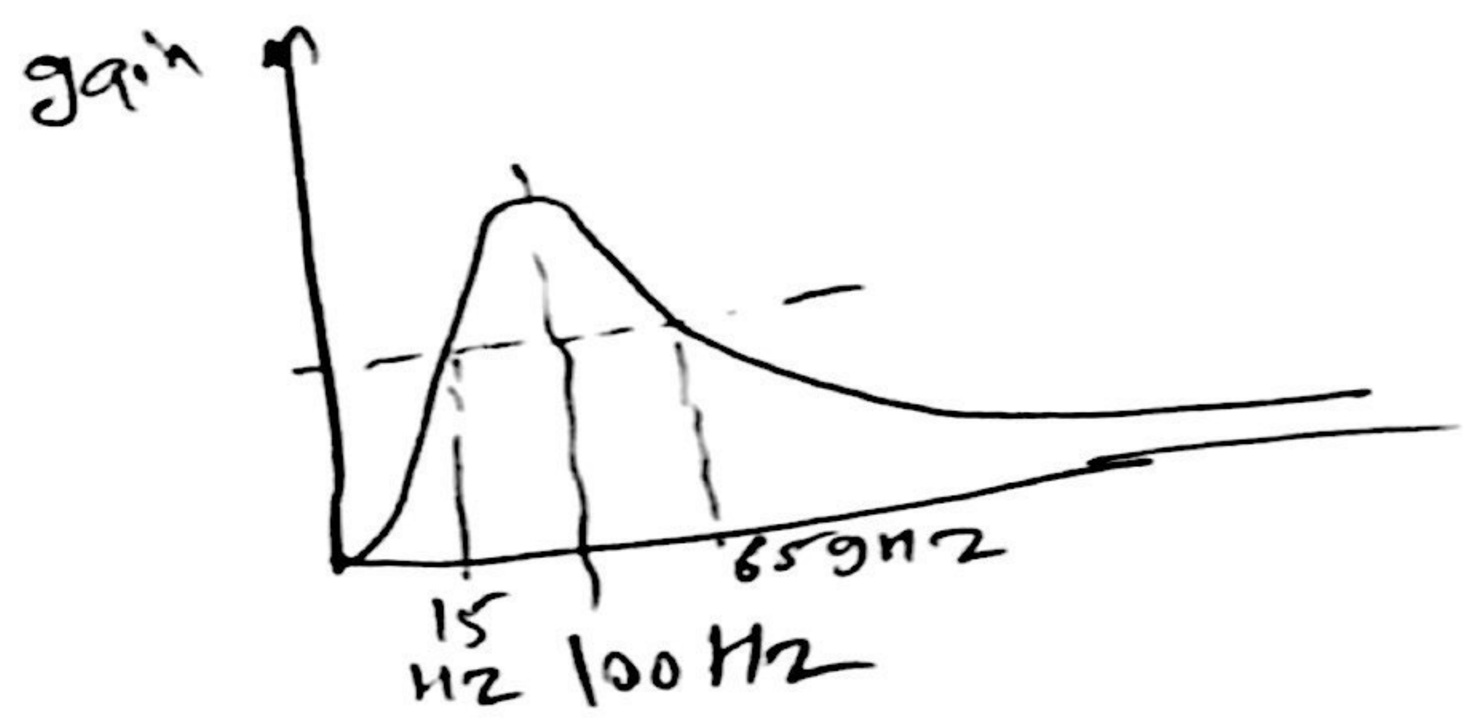
$f_1 = \frac{\omega_1}{2\pi} = \boxed{659\text{ Hz.}}$ cut-off

Case 2: $2.53\times10^{-5}\omega - \frac{10}{\omega} = -0.1025$

$\Rightarrow$ solve for ω

$\Rightarrow$ $\omega_2 = 95$ rad/sec.

$f_2 = \boxed{15\text{ Hz}}$ cut-off



this design will filter 1000 Hz siren effectively but may not filter 20 Hz noise effectively as it is within it's cut-off freq.